

# Antonio Ferrara

AI Researcher

Florence, Italy

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AI researcher with a PhD in algorithmic fairness, working on LLM alignment and bias evaluation, grounded in statistical testing. Developing methods to audit and mitigate bias in AI systems. Published at NeurIPS, KDD, and VLDB.

## Research

### Selected Publications

- **Ferrara, A.**, Cozzi, F., Perotti, A., Panisson, A., & Bonchi, F. (2025). Size-adaptive Hypothesis Testing for Fairness. *Advances in Neural Information Processing Systems*, 38, 97935–97971. **(NeurIPS 2025)**
- **Ferrara, A.**, García-Soriano, D., & Bonchi, F. (2025). Beyond Shortest Paths: Node Fairness in Route Recommendation. *Proceedings of the VLDB Endowment*, 18(9), 3230–3242. **(VLDB 2025)**
- **Ferrara, A.**, Bonchi, F., Fabbri, F., Karimi, F., & Wagner, C. (2024). Bias-aware ranking from pairwise comparisons. *Data Mining and Knowledge Discovery*, 38(4), 2062–2086. **(DAMI 2024)**
- **Ferrara, A.**, Espín-Noboa, L., Karimi, F., & Wagner, C. (2022). Link recommendations: Their impact on network structure and minorities. *14th ACM Web Science Conference*, 228–238. **(WebSci 2022)**
- **Ferrara, A.\***, Pugnana, A.\*, Bonchi, F., & Ruggieri, S. (2026). Bounded-Abstention Pairwise Learning to Rank. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. **(KDD 2026, to appear)**
- **Ferrara, A.**, Abrate, C., Vitale, F., & Bonchi, F. (2026). Auditing for Demographic Bias in Opaque Rankings. *Proceedings of the VLDB Endowment*, 19. **(VLDB 2026, to appear)**

**Reviewing** NeurIPS, ICLR, KDD, ICML, ACL '25 · NeurIPS, EWAF '24 · ACM TSC '23 · WWW '22 · ACM HT '21 · AAAI '25 (PC)

## Work Experience

### Intesa Sanpaolo

AI Researcher

Italy

Jan. 2026 – Present

- Developing a bias-aware variant of DPO for fine-tuning and debiasing large language models.
- Building an LLM-based thematic stock basket construction system using semantic coherence scoring.
- Prototyping a GDPR-compliant GenAI pipeline for music generation.

### CENTAI (acquired by Intesa Sanpaolo)

AI Researcher

Italy

Oct. 2023 – Jan. 2026

- Evaluated LLM-as-a-judge for bias in automated CV screening.
- Built a churn prediction model using RNNs for a major digital bank.
- Developed a clustering pipeline for financial asset valuation.
- Audited AI models for disparities across patient groups in breast cancer radiotherapy (Horizon Europe).

### GESIS – Leibniz Institute for Social Sciences

Early Stage Researcher

Germany

Oct. 2020 – Sep. 2023

- Researched fair ranking algorithms and bias in network-based recommendations.

### EURECAT – Technology Centre of Catalonia

Internship

Spain

Jan. 2023 – May 2023

- Developed a bias-aware ranking method from pairwise human feedback.

### European Central Bank

Traineeship

Germany

May 2020 – Sep. 2020

- Monitored a large-scale Credit Risk Parameters database.
- Built a machine learning model to detect outliers in crypto-assets trading platforms.

### ISI Foundation – Institute for Scientific Interchange

Internship

Italy

Sep. 2019 – Apr. 2020

- Researched connections between Graph Neural Networks and spatial statistics.

## Education

### Graz University of Technology

PhD in Computer Science

Austria

Oct. 2020 – Jun. 2026

- Final grade: Sehr gut (1), the highest grade on the Austrian scale.
- Supervisor: Prof. Dr. Fariba Karimi.
- Title: Statistical and Structural Approaches to Algorithmic Fairness.

## European Union's Horizon 2020 (Marie Skłodowska-Curie grant)

NoBIAS – Artificial Intelligence Without Bias ITN

- Marie Curie ITN training 15 researchers on bias detection and fairness in AI systems.

Germany

Oct. 2020 – Sep. 2023

## University of Turin

Master's Degree in Stochastics and Data Science

- Final grade: 110/110 cum Laude.
- Thesis: Thinking in space: a comparison between spatial statistics and graph neural networks.

Italy

Sep. 2017 – Apr. 2020

## Johannes Kepler University

Erasmus+

Austria

Sep. 2018 – Jul. 2019

## University of Florence

Bachelor's Degree in Mathematics

- Final grade: 110/110.
- Thesis: Cryptographic applications of the LLL algorithm.

Italy

Sep. 2013 – Oct. 2017

## Honours and Awards

2023 **€220K DFG Grant**, Bias in ranking systems – University of Mannheim (declined to join CENTAI)

Germany

2025 **1st Place**, DRW Trading Challenge – NeurIPS 2025

USA

## Skills

**Research:** Algorithmic Fairness · Bias Detection & Mitigation · Statistical Hypothesis Testing · LLM Evaluation & Alignment · RLHF/DPO · Ranking & Recommender Systems · Deep Learning

**Engineering:** Python · PyTorch · Hugging Face/Transformers · R · Git · Shell/Unix · SQL

**Languages:** Italian (native) · English (C2, full professional) · Spanish (B2) · German (A2)